

$^1\text{H}(^{36}\text{Ar},\text{d})$ [2010Le03,2011Le01](#)

$J^\pi=0^+$ for ^{36}Ar ground state.

[2010Le03](#), [2011Le01](#): A ^{36}Ar beam at 33 MeV/u was provided at the National Superconducting Cyclotron Laboratory, MSU.

Targets were polyethylene (CH_2)_n. Deuterons were detected using the High-Resolution Array (HiRA) of Si and CsI(Tl) telescope detectors in coincidence with recoil residues identified in the S800 spectrometer by the focal plane ionization chamber andToF.

Measured $\sigma(E_d, \theta)$ in inverse kinematics. Deduced neutron spectroscopic factors from adiabatic distorted wave approximation (ADWA) analysis of the measured $\sigma(\theta)$ using Chapel-Hill global optical potential parameters (CH89) and JLM optical potentials and geometry for transferred neutron constrained by Hartree–Fock calculations (JLM+HF). Comparisons with shell–model calculated spectroscopic factors.

Theoretical studies involving $^1\text{H}(^{36}\text{Ar},\text{d})^{35}\text{Ar}$: [2011Nu01](#), [2023He15](#).

 ^{35}Ar Levels

<u>E(level)[†]</u>	<u>J^π</u>	<u>$L^\ddagger$</u>	<u>$C^2S^\ddagger$</u>	<u>Comments</u>
0	$3/2^+$	2	2.3 2	C ² S: others: 1.6 <i>I</i> from 2011Le01 using JLM+HF calculations; 2.10 from large basis–shell model calculations (2010Le03); 2.21 <i>49</i> from a reanalysis of the $\sigma(\theta)$ data using finite-range ADWA (2011Nu01), including theoretical uncertainties associated with optical potentials (7%) and the approximate solution of three-body problems (19%); 2.1 +2–4 from a reanalysis of the $\sigma(\theta)$ data using ADWA within a Bayesian framework (2023He15), including theoretical uncertainties associated with optical potentials.
1180		0	1.2 <i>I</i>	
2980 [#]				
3190 [#]				
5570				

[†] From [2010Le03](#).

[‡] From ADWA analysis of measured $\sigma(\theta)$ in [2011Le01](#) using CH89 optical potential parameters.

[#] Doublet in measured spectra.